

English
Medium



EDU TERIA

72nd BPSC

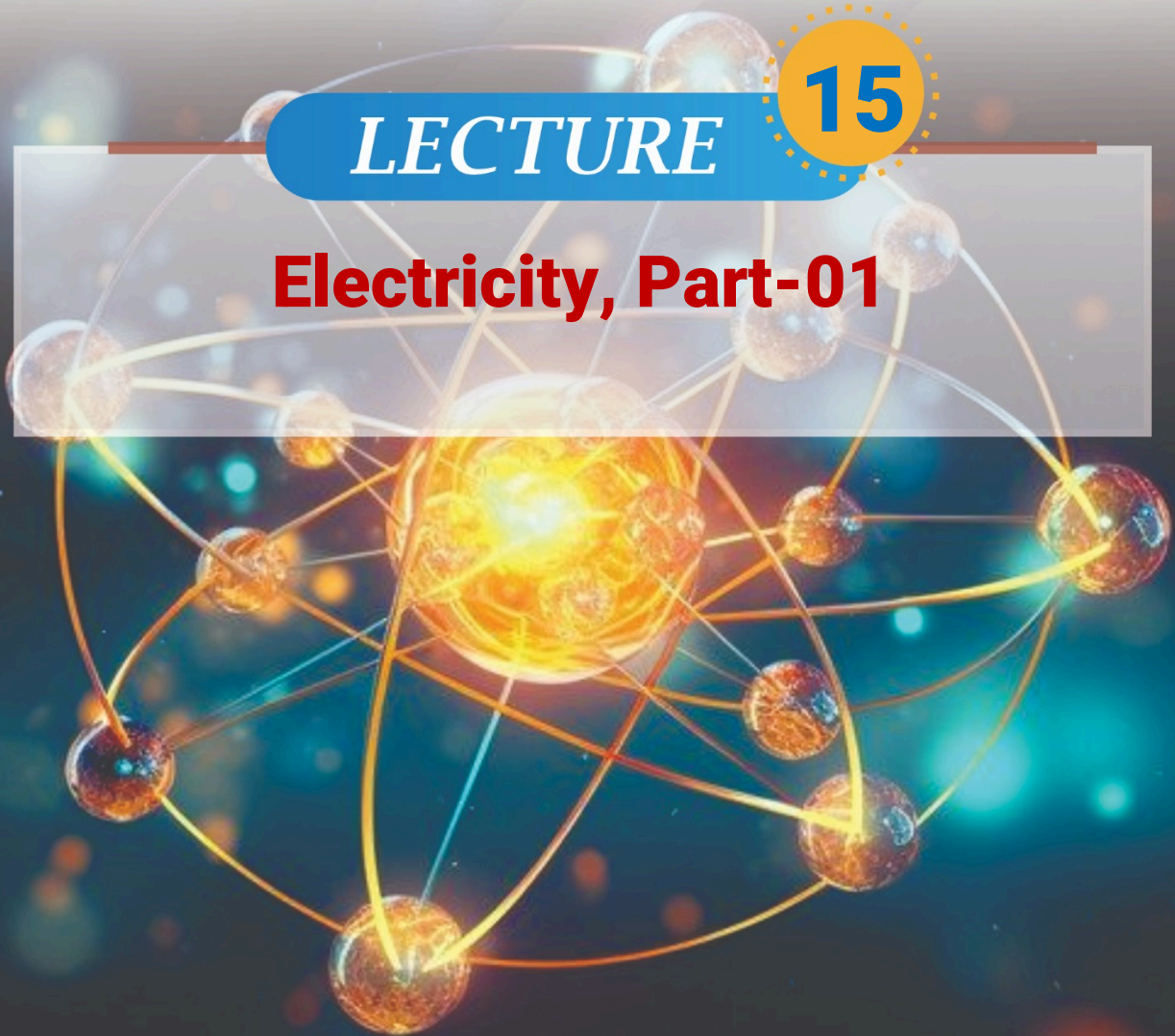
Prelims Online Batch

PHYSICS

LECTURE

15

Electricity, Part-01

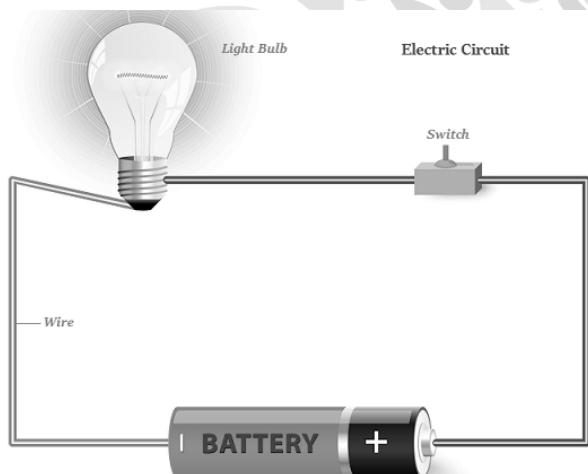


Electricity, Part-01

Electric charge is the fundamental property of matter responsible for electrical phenomena. It can be positive (+) or negative (-) and is conserved. Charge is the property associated with matter due to which it produces and experiences electrical and magnetic effects. The excess or deficiency of electrons in a body is the causes of net charge on a body.

- Unit: Coulomb (C)
- Nature: Scalar
- Charge on one electron = 1.6×10^{-19}

Note : Charge of fundamental particles (i.e. electron, proton etc.) is their internal characteristic while charge on a body depends on the number of protons & electrons inside the body.



Types of Electric Charge (q):

- Positive (+)
- Negative (-)

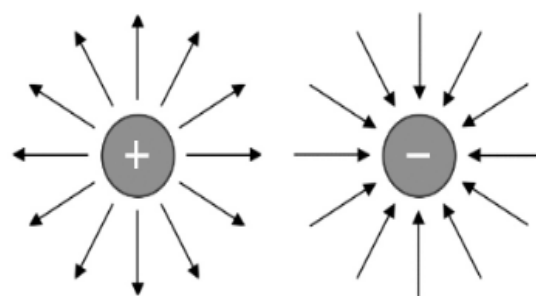
Conservation of Charge: Total charge remains constant

Electric Field

(i) Same charge repel each-other



(ii) Opposite charges attract each other.



Positive charge

Negative charge

Specific Properties of Charge

- Charge is a scalar quantity : It represents excess or deficiency of electrons.
 - Charge is transferable : If a charged body is put in contact with an another body, then charge can be transferred to another body.
 - Charge is always associated with mass
 - Charge cannot exist without mass though mass can exist without charge.
 - So the presence of charge itself is a convincing proof of existence of mass.
 - The mass of a body changes after being charged.
 - When a body is given a positive charge, its mass decreases.
 - When a body is given a negative charge, its mass increases.
 - Charge is quantised
 - The quantization of electric charge is the property by virtue of which all free charges are integral multiple of a basic unit of charge represented by e. Thus charge q of a body is always given by $q = ne$ n = positive or negative integer
 - The quantum of charge is the charge that an electron or proton carries.
- Note :** Charge on a proton = (+) charge on an electron = 1.6×10^{-19} C
- Charge is conserved
 - In an isolated system, total charge does not change with time though individual charge may change, i.e. charge can neither be created nor destroyed. Conservation of charge is also found to hold good in all types of reactions either chemical (atomic) or nuclear. No exceptions to the rule have ever been found.
 - Charge is invariant





- Charge is independent of frame of reference. i.e. charge on a body does not change whatever be its speed

Coulomb's Law

- The electrostatic force of interaction between two static point electric charges is directly proportional to the product of the charges, inversely proportional to the square of the distance between them and acts along the straight line joining the two charges.
 - Consider two point charges q_1 and q_2 separated by a distance r . Let F be the electrostatic force between these two charges. According to Coulomb's law,
 - Force between two charges:
 - Like charges repel, unlike attract
 - Force decreases with distance
- $F \propto q_1 q_2$
 - $F \propto 1/r^2$

After combining

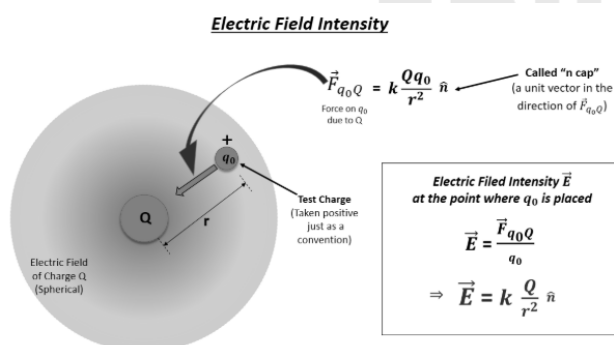
$$F = k \frac{q_1 q_2}{r^2} = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}$$

Electric Field Intensity (E)

- Defined as force per unit charge

$$E = \frac{F}{q} = \frac{\text{Force}}{\text{Charge}}$$

- Unit:** N/C
- Direction:** Force on positive charge



- Work done per unit charge

$$V = \frac{W}{q}$$

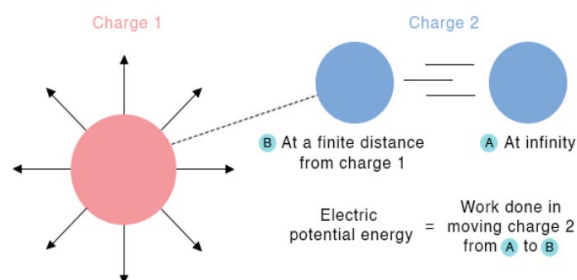
- Unit:** Volt (V)

Potential Difference

Potential difference (voltage) is the difference in electrical potential energy per unit charge between two points in a circuit, driving current flow.

- Difference in potential between two points

- Drives current in a circuit



Electric Current (I)

An electric current is a flow of charged particles, such as electrons or ions, moving through an electrical conductor or space. It is defined as the net rate of flow of electric charge through a surface.

- Flow of electric charge $I = \frac{Q}{t}$
- Unit:** Ampere (A)
- Direction:** Conventional (+ to -)

